

Woolworths Online Order and Delivery System – Enterprise Systems Analysis, Design, and Modelling

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Organisational and System Analysis

Problem Statement

As online shopping continues to grow in popularity due to its convenience, Woolworths New Zealand has identified a need to implement an online system of its own (this is an assumption that Woolworths New Zealand does not have an online ordering and delivery system; the proposed idea is based on this assumption). Currently, Woolworths New Zealand does not have a dedicated online order and delivery system, which is a significant issue for multiple departments to perform effectively.

Customer service challenges

From the customer's perspective on the manual purchasing of groceries, there is no efficient way in which customers are able to compare and decide different product choices based on their nutritional value, price, and other product attributes. Customers would also not be able to access Woolworths online, as there is no system that provides the ability for customers to purchase goods. The customer would not have the convenience or mobility issues to be able to travel to a supermarket.

From the standpoint of a customer service representative, the absence of an online platform implies that customer service would have to be conducted in person, in-store for refunds, and deal with customer complaints, as well as have no organised method to manage digital customer complaints. This makes it difficult for Woolworths New Zealand to quickly address, track, and solve issues. This would erode consumer loyalty and confidence.

Operational challenges

The challenges that the store operation employees faced were that they were not able to visualize the total number of products offered in the store, thus needing to manually record and manage product stocks, which could increase the chances of human error. Employees would not be able to have

statistical data on which product has the most demand, which leads to poor positioning of where products should be stocked.

Shift managers have no digital tools to delegate tasks or monitor the progress of staff, relying entirely on in-person communication. Store managers have no central system or efficient way for the store manager to track the performance of the store, which would help determine potential operational bottlenecks. Currently, the store manager relies on manual reporting, which is time-consuming and a possible task that could involve human error.

Supply chain and logistics impact

This issue would also affect the supply chain and logistics department, as procurement specialists are unable to make the most effective decisions in purchasing and sourcing products due to a lack of live visibility on real-time demand data. This provides inaccurate data to the procurement specialist, which would also affect the inventory controller, as they monitor and maintain optimal stock levels across store locations based on the procurement specialist's data. This would risk overstocking or understocking across various locations.

This also impacts delivery drivers, as ineffective supply allocation across store locations makes it difficult to plan and fulfil deliveries efficiently, raising overall operating expenses. Delivery drivers have no structured way to receive or manage their delivery schedules, as these are assigned manually without consideration of the driver's current location or availability. This results in inefficient route planning and delays in order fulfilment. Furthermore, delivery drivers have no way of indicating the status of the product delivery to the customers or indicating failed deliveries.

Proposed Solution

Woolworths New Zealand needs to develop an Online Order and Delivery System that allows customers to place an online order, in which customers are allowed to browse and search for products online. The customer should also be able to add products to their shopping cart and be allowed to check out. Customers would also be able to save a shopping list to reuse previous lists of orders to

make it faster to shop for weekly used products. Customers will be able to create an account so that their details and saved shopping list are stored on the account. Customer service representatives should have access to a digital complaint and inquiry management system that allows them to log, track, and resolve customer issues efficiently, improving response times and maintaining customer loyalty.

In the store operation department, the store operation employee would be able to view digital information on the online system about what orders to pick and pack to prepare for delivery. The Shift Managers would use the system to delegate tasks to the store operation employees and be able to monitor their progress. The store manager would be able to have a view of performance metrics (tracking orders received, processed, and fulfilled), which would help identify bottlenecks or delays. Other task that is needed to be able to be performed include inventory oversight, reviewing customer feedback, and managing store information (display of online promotions, announcements, and opening hours).

For the supply chain and logistics department, the system should provide procurement specialists with access to real-time demand data, receiving messages about stock levels, especially if they are low, and track purchases made and the delivery of suppliers on the system. The procurement specialist should also be able to generate reports for planning future purchases. Inventory controllers should be able to monitor live stock levels across all store locations through the system, and monitor incoming stocks to ensure inventory is correctly recorded. Just like the procurement specialist, the inventory controller would also be able to generate reports on inventory to see which products need restocking the most and the least.

Delivery drivers should be assigned schedules automatically based on their availability and location, and should be able to update delivery statuses in real time so that customers are notified of their order progress. Drivers should also be able to flag issues on the system that need urgency.

Organisation Overview

Woolworths New Zealand (rebranded from Countdown) is a subsidiary of Woolworths Group Limited, an Australian publicly listed company. Woolworths New Zealand is a major supermarket chain whose headquarters are located in Mangere, Manukau, Auckland. Woolworth Group was founded by Percy Christmas on September 22, 1924, in Sydney, Australia. It is one of the 3 major supermarkets (Pak'n Save, Woolworths, New World) in New Zealand. Currently, there are 185 supermarkets^[1] in New Zealand and 20,000 employees^[1] working in New Zealand.

Woolworth's core values are providing convenience, value, range, and quality, and a motto of "The Fresh Food People"^[2]. Thus, Woolworths needs dependable information systems to manage its operations across shops, distribution centers, and digital platforms because it depends greatly on customer satisfaction. One of these operations that would be focused on in this report is the Online and Delivery System, since it offers greater convenience and the ability to save time by ordering groceries online. These advantages in these systems align with the organisation's values, which are to provide the best experience for customers.

Departmental Structure and Responsibilities

In the ordering and delivery systems in Woolworths New Zealand, there are 4 critical departments in which forms the operation of the system. These systems are Store Operations, Supply Chain & Logistics, Customer Service, and IT.

- Store Operations

The store operations department for Woolworths New Zealand in 185 supermarkets^[1] focuses on restocking shelves and maintaining the food freshness of merchandise. In the context of an online ordering and delivery system, they are responsible for managing orders online by picking and packing them for delivery.

- Supply Chain & Logistics

The Supply Chain & Logistics department for Woolworths New Zealand focuses on managing stock movement from the supplier and distribution centres to the store. In the context of an online ordering and delivery system, they are responsible for assigning deliveries to complete orders and managing their delivery schedule, as 22%^[3] of orders online are fulfilled within 2 hours, and 57%^[3] of orders online are fulfilled within the same day of ordering, as well as ensuring there is enough stock for deliveries.

- Customer Service

The Customer Service department for Woolworths New Zealand focuses mainly on making sure that all customers are satisfied with the service and merchandise by handling customer inquiries, complaints, and issues. In the context of an online ordering and delivery system, this department is responsible for handling customer complaints and issues, assisting customers online, and processing refunds for incorrect or out-of-stock merchandise on the online ordering and delivery system.

- IT

The IT department for Woolworths New Zealand focuses on maintaining all digital platforms and technology systems, ensuring that user data is secure, developing improvements to the website, and resolving technical issues. In terms of the context of the online ordering and delivery system, the IT department is responsible for maintaining the system, integrating inventory, supply chain, and payment systems.

- People and Culture (HR)

The People and Culture Department for Woolworths New Zealand focuses on making sure that all employees are well-trained, supported in their well-being, and manage payroll and recruitment. In the 2025 annual report, mental health programs such as 'Sonder' and leadership skills are completed by over 55,000^[3] employees. This department also manages wage agreements and ensures there is an inclusive environment and culture.

- Finance

The finance department for Woolworths New Zealand is involved in managing the budget, managing the cash flow, planning, and analysing financial decisions, analysing financial risk, as well as producing financial reports. These responsibilities are important in overseeing the financial report of Woolworths New Zealand. Food's total sales reached \$8,286 million^[3] with normalised sales.

- Marketing

The marketing department for Woolworths New Zealand's main goal is to build brand awareness, build brand loyalty through rewards, and research the trends of the market, creating promotions and advertisements. This includes the marketing department managing the grocery shopping website, which contributed to having 2.1 million^[3] active everyday reward members in Q4 of 2025.

Stakeholders and User Roles

Department	Stakeholder	Role
Store Operations	Store Operations Staff	Picks and prepares orders in the store
Store Operations	Store Manager	Oversees overall store operations and monitors system performance.
Store Operations	Shift Manager	Oversees store operations and staff delegation

Supply Chain and Logistics	Delivery Driver/Employee	Fulfil deliveries to customers
Supply Chain and Logistics	Procurement Specialist	Specialist Purchases stock for store locations
Supply Chain & Logistics	Inventory Controller	Manages stock distribution across store locations
Customer Service	Customer	Primary user of the online shopping platform
Customer Service	Customer Service Representative	Handles customer complaints and refunds

Information from Stakeholders

The methods used to gather information from stakeholders are mainly interviewing stakeholders with questions. As well as viewing similar systems in the same industry to find out what the existing common features are in the system.

Customer stakeholder questions (This response is from a friend acting as a customer stakeholder):

Question	Answer
What are some of the problems you have with the ordering process in existing systems from other organisations?	When ordering something, finding the product is hard. Sometimes the filter does not work. Adding items to the cart is slow. Item stock is not displayed properly (want real-time stock information).

How often do you currently shop for groceries, and what frustrates you most about the in-store experience?	The client said he went shopping once a week and that the thing that frustrates him the most is the checkout lines, which take too long.
What would make you choose online grocery shopping over visiting a store in person?	Prices should not be too expensive for online ordering, and people selecting items should be choosing quality products. The online experience should be nice and easy to use, such as not having too many unnecessary buttons (search bar, filter).
How would you prefer to be notified about your delivery status – SMS, email, or in-app notifications?	In app notifications, sms should be an option as another method of notifying.
Would access to exclusive online-only deals or Everyday Rewards points influence how often you shop online?	The stakeholder stated that if money is saved, filling out a form is not required. Then a deal or an everyday reward feature would be fine to have.

Shift Manager stakeholder questions (This response is from me as the stakeholder):

Question	Answer
How do you currently assign tasks to store operation staff, and what are the biggest challenges with this process?	Currently, the shift manager assigns tasks, having scheduled meetings daily on what each staff member's task is to complete, and the biggest challenge is having to organise a in person meet everytime since it is time-consuming.

How do you currently monitor the progress of staff during a shift, and how time-consuming is this?	To monitor the staff's progress, each staff will have a paper time sheet for each staff member, and when they are done with their shifts, they write down on that sheet. The sheets are collected at the end of every day to be evaluated by the shift manager.
What are the most common issues that arise during a shift that slow down operations?	Common issues are that it is hard to make any changes in plans, and you will need to meet in person to inform employees.
How are staff currently informed about which products need to be picked and packed for an order?	Staff are informed by getting a paper receipt on the physical board. When they pick and pack based on the first-come, first-served method.
How do you currently handle situations where a staff member falls behind on their tasks or is unavailable?	Additionally, help would be assigned to the task that is not up to schedule. Assess the situation for any improvements that could be made.
What features would be most useful to you in a system that allows you to delegate and monitor tasks digitally?	The feature where I could see all the store operation staff, tasks, and time spent towards the goal, as well as tracking the progress of each of them.

How would you prefer to view the progress of tasks – through a dashboard, notifications, or another method?	Prefer to view them through a dashboard with all the important features, metrics, and notifications when there is urgent information.
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Procurement Specialist stakeholder questions (This response is from me as the stakeholder):

Question	Answer
What are the biggest challenges you face when sourcing and ordering products?	Identify which supplier is the most suitable for the budget and standards. As researching suppliers is done manually.
How do you currently track whether a supplier has fulfilled a purchase order?	This is done by inquiring with the supplier about their progress in fulfilling the order and tracking them on a spreadsheet to be checked manually on a daily basis.
How do you currently forecast how much stock needs to be ordered for each product?	Stock is forecast based on historical data, and purchasing decisions are made based on the sales record of the past rather than in real time.
How do you currently share procurement information with the inventory controller?	Once a supplier is selected, the information about the stock is written on a spreadsheet and shared with the inventory controller so that the inventory controller can manage the incoming stock.
What information would you need real-time access to in order to make better purchasing decisions?	Inventory, order tracking, real-time demand, Market & Pricing Data

How do you know when stocks are low and are in need of purchasing products from suppliers?	A spreadsheet is used to track the amount of stock currently in store daily. Purchasing decision is only made after stocks are needed according to the daily spreadsheet that records the inventory.
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System scope, boundaries, and assumptions

Scope

The system consists of two sides: the customer side, where it is the online ordering and delivery platform for customers, and the backend side, the internal management system that supports the operations of the staff. Three departments span the system, which include store operations, customer service, and supply chain & logistics. Each department would have varying levels of access through the internal management system.

Boundaries

In scope:

- Online ordering and delivery platform
- Internal staff management system across Store Operations, Supply Chain and Logistics, and Customer Service departments
- Real-time stock and inventory management
- Automated delivery scheduling and tracking
- Digital complaint and inquiry management

Out of scope:

- Payroll
- HR management
- financial accounting
- in-store sales

- supplier direct access to the system
- marketing and advertising

Assumptions

1. Assuming Woolworths has not implemented an order and delivery system and still relies on in-person grocery shopping.
2. Assuming standard industry benchmarks for staff roles, as I have no visibility into Woolworths' specific operating procedures.
3. All staff are assumed to have access to a device to use the internal management system.
4. Stock levels in the system are assumed to be accurate and up to date.
5. Payment is assumed not to be handled by the system and is processed through a 3rd party payment gateway.

Requirements

Business Goals

Customer Service

- Improve customer convenience by providing an accessible online grocery shopping platform.
- Enhance customer satisfaction through effective delivery communication.
- Improve customer trust through an accessible online complaint and refund platform.

Store Operations

- Decrease the amount of in-person communication that shift managers and retail employees use.
- Reduce human error in store operations through digital task and order management.
- Increase stock management precision with real-time stock visibility.

Supply Chain & Logistics

- Use automated delivery scheduling to increase delivery efficiency.
- Incorporate real-time delivery status updates on deliveries.
- Improve forecasting accuracy by providing procurement and inventory staff with real-time data demands.

User Requirements

Customer should be able to:

- Browse and search for products online.
- Compare products based on nutritional value and price.
- Add products to a shopping cart and check out.
- Save a shopping list for reuse.
- Create an account to store their details and saved lists.
- View store promotions, opening/closing dates, and news.
- Track and receive notifications about their delivery status.
- Choose delivery options / pick up.

A customer service representative should be able to:

- Logging and managing customer complaints.
- Processing refunds for incorrect or out-of-stock items.
- Log and manage customer complaints digitally.

Store Operation Staff should be able to:

- View real-time stock and product inventory.
- Receive order information to pick and prepare orders for delivery.

Shift Manager should be able to:

- Delegate tasks to store operations staff through the system.
- Monitor the progress of staff tasks in real time.

Store Manager should be able to:

- Track overall performance.
- Track order fulfilment rates.
- Identify bottlenecks from analysing overall store operations.
- Oversee inventory levels in the store.

A delivery driver should be able to:

- Get automatically assigned delivery orders based on location and availability.
- Update the delivery schedule when assigned to a delivery.
- Flag urgent delivery issue.

A procurement specialist should be able to:

- Access real-time demand data to make informed purchasing decisions.
- Receive automated alerts when stock levels are low.
- Track purchases made and supplier deliveries through the system.
- Generate reports for planning future purchases.

An inventory controller should be able to:

- Monitor real-time stock levels across all store locations.
- Monitor incoming stock to ensure inventory is correctly recorded.
- Generate inventory reports to identify which products need restocking the most and least.

Functional Requirements

Customer

- The system must allow customers to browse and search for products by name, category, price, and nutritional value.
- The system must allow customers to add products to a shopping cart.
- The system must allow customers to remove products from a shopping cart.

- The system must allow customers to proceed to checkout from their shopping cart, where they can review their orders.
- The system must allow customers to select their delivery information (time, destination) and submit their order.
- The system must allow customers to create a personal account with their name, email address, password, and delivery address.
- The system must allow customers to save shopping lists from previous orders for reuse.
- The system must display the current delivery status of an order to customers, including order confirmed, order being prepared, out for delivery, and delivered.
- The system must automatically send delivery notifications to customers when their delivery status is updated.

Customer Service Representative

- The system must allow customer service representatives to log customer complaints and inquiries digitally.
- The system must allow customer service representatives to track the status of open complaints and inquiries.
- The system must allow customer service representatives to update and resolve logged complaints.
- The system must allow customer service representatives to process refunds for incorrect or out-of-stock items.
- The system must allow customer service representatives to send notifications to customers via in-app notifications or SMS regarding issues with their order.

Store Operation Staff

- The system must display the list of products and their quantities for each incoming order to store operations staff.
- The system must display the order priority status to store operations staff so that urgent orders are fulfilled first.
- The system must display the delivery deadline for each order to the store operations staff.

- The system must allow store operations staff to update the status of an order as it is being picked and packed.
- The system must display real-time stock levels and product inventory to store operations staff.
- The system must send a notification to store operations staff when a new order is assigned and requires picking and packing.

Shift Manager

- The system must allow shift managers to assign tasks to store operations staff.
- The system must allow shift managers to monitor the progress of assigned tasks in real time through a dashboard.
- The system must send a notification to the shift manager when an assigned task has not been updated within a defined time period.
- The system must allow shift managers to reassign tasks to different staff members when needed.

Store Manager

- The system must allow the store manager to track the orders received.
- The system must allow the store manager to view the orders that are being processed.
- The system must allow the store manager to view the fulfilled orders.
- The system must allow the store manager to view the stock quantities for each product in the store.
- The system must allow the store manager to review customer feedback that is submitted through the system.
- The system must allow the store manager to update in-store information on the online platform, which includes online promotions, announcements, and opening hours.

Procurement Specialist

- The system must allow the procurement specialist to generate reports, including current stock levels, product demand trends, and outstanding purchase orders.

- The system must notify the procurement specialist on the online platform when the stock is below a defined threshold level. The notification must apply to all the store locations that the procurement specialist is assigned to.
- The system must allow the procurement specialist to create and submit purchase orders to suppliers and track their delivery status until fulfilment.
- The system must allow the procurement specialist to view product inventory.
- The system must provide procurement specialists with access to supplier market and pricing data, enabling them to compare suppliers based on cost, quality standards, and reliability.

Inventory Controller

- The system must allow the inventory controller to view product inventory across assigned store locations on the online platform.
- The system must allow the inventory controller to monitor incoming stocks of suppliers, automatically updating inventory records upon delivery confirmation to ensure stock counts remain accurate across all locations.
- The system must allow inventory controllers to generate inventory reports that rank products by restocking frequency and current stock levels.

Delivery employee

- The system must automatically generate a delivery schedule based on the delivery driver's location and availability.
- The system must allow the delivery driver to update their status to dispatched, in transit, and delivered.
- The system must allow the delivery driver to flag urgent issues on the system if a delivery has any issues, such as failed deliveries, incorrect stock, or vehicle problems. This issue is then notified to the appropriate department.

Non-Functional Requirements

Product requirements

- The system must load product search results within a 5-second to ensure a smooth browsing experience on the ordering platform.
- The system should have at most 1 minute of delay in how long it takes for the stock to update on the ordering platform.
- The system for customers should be 4 or fewer steps away from ordering a product to the checkout.
- The system must be accessible on both mobile and desktop devices for the convenience of the customer.
- The system must securely store customer personal data, including name, email address, password, and delivery address.
- The system must be able to handle millions of concurrent users.
- The system must restrict each user role so that staff only access features relevant to their department.

Organisational requirements

- The system must be implemented using c sharp as it is stated that this is the development language requirement.
- The system must be in line with Woolworths New Zealand's privacy guidelines^[4].
- The system must integrate with a third-party payment gateway to process all customer payments.

External requirements

- The system must comply with the New Zealand Privacy Act 2020 when storing customer data.

Traceability between stakeholders and requirements

Stakeholder	Source	Requirement	Justification
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Customer	Stakeholder Interview & Problem Statement & Legacy System	The system must allow customers to browse and search for products by name, category, price, and nutritional value.	The stakeholder stated that, in general, many different online ordering websites find it hard to operate, and filters do not always work. In the problem statement, customers have no efficient way to compare products based on nutritional value, price, and other attributes. Observation of existing online grocery platforms, such as PaknSave, has filter functionality to make it easier to browse wanted products.
Customer	Problem Statement & Legacy System	The system must allow customers to add products to a shopping cart.	The problem statement identifies that customers have no ability to purchase goods online. Observing from existing similar system there needs to be a cart section where customers can view all the products that they have selected.
Customer	Problem Statement & Legacy System	The system must allow customers to remove products from a shopping cart.	The problem statement identifies that customers have no ability to manage their grocery purchases online. Managing includes removing products from the cart. This is standard functionality when observing similar online ordering and delivery systems in the same industry.
Customer	Problem Statement & Legacy System	The system must allow customers to proceed to checkout from their shopping cart, where they can review their orders.	As stated in the problem statement identifies that customers cannot purchase goods online. As well as observing the existing system, customers would be able to proceed with their checkout with payment and than delievery.

Customer	Customer Stakeholder Interview	The system must allow customers to select their delivery information (time, destination) and submit their order.	The customer stakeholder stated that the delivery price should be affordable for online ordering and that the online experience should be easy to use.
Customer	Problem Statement & Legacy System	The system must allow customers to create a personal account with their name, email address, password, and delivery address.	The problem statement identifies that customers have no ability to access Woolworths online or store their details. Observation of existing online grocery platforms confirmed that account creation with name, email, and password is a common feature to create an account.
Customer	Problem Statement	The system must allow customers to save shopping lists from previous orders for reuse.	The problem statement identifies that customers have no convenient way to manage their grocery purchases online. Saving shopping lists addresses the need for customers to quickly reorder weekly used products without having to search for them again each time.
Customer	Customer Stakeholder Interview & Legacy System	The system must display the current delivery status of an order to customers, including order confirmed, order being prepared, out for delivery, and delivered.	This requirement is based on the customer stakeholder interview, as they stated that they prefer in-app notifications and SMS as a method of being notified about their delivery status. The requirement is to expand on this concept about the stages of notification to the customer. This feature is seen in online grocery platforms, which display multiple stages of order progress to customers.
Customer	Customer	The system must	The Customer stakeholder stated they

	Stakeholder Interview	automatically send delivery notifications to customers when their delivery status is updated.	prefer in-app notifications with SMS as an alternative method of notifying them of their delivery status.
Customer Service Representative	Problem Statement	The system must allow customer service representatives to log customer complaints and inquiries digitally.	The problem statement identifies that customer service has no organised method to manage digital complaints. This makes it difficult for Woolworths New Zealand to manage customer complaints. Therefore, an online platform functionality is needed to store and manage this efficiently.
Customer Service Representative	Problem Statement	The system must allow customer service representatives to track the status of open complaints and inquiries.	In the problem statement, Woolworths New Zealand has determined that there is difficulty in quickly addressing and track issues. Thus, it is required that the system track these complaints so that it could be solved in the most effective manner.
Customer Service Representative	Problem Statement	The system must allow customer service representatives to update and resolve logged complaints.	The problem statement identifies that customer service has no organised method to manage complaints, indicating a need for a system that allows representatives to update and resolve logged complaints efficiently to improve response times.
Customer Service Representative	Problem Statement	The system must allow customer service representatives to process refunds for incorrect or out-of-stock items.	The problem statement identifies that refunds are currently handled in person in-store, making it difficult to process them efficiently.

Customer Service Representative	Customer Stakeholder Interview	The system must allow customer service representatives to send notifications to customers via in-app notifications or SMS regarding issues with their order.	The customer stated they prefer in-app notifications with SMS as an alternative method of notification when there are issues with the order.
Store Operations Staff	Problem Statement & Shift Manager Stakeholder Interview	The system must display the list of products and their quantities for each incoming order to store operations staff.	The problem statement identifies that staff cannot visualise the total products and rely on paper receipts. In the shift manager stakeholder interview, the stakeholder stated that staff are currently informed about orders through paper receipts on a physical board. Therefore, this requirement is needed for the convenience of store operation staff.
Store Operations Staff	Shift Manager Stakeholder Interview	The system must display the order priority status to store operations staff so that urgent orders are fulfilled first.	The shift manager stated that staff currently pick and pack based on a first-come, first-served method from a physical board, with no priority system in place.
Store Operations Staff	Problem Statement & Shift Manager Stakeholder Interview	The system must display the delivery deadline for each order to the store operations staff.	Staff have identified that there are no digital tools to manage order preparation efficiently. The shift manager stakeholder stated that staff currently rely on paper receipts with no indication of delivery deadlines.
Store Operations Staff	Shift Manager Stakeholder	The system must allow store operations staff to update the	The shift manager stated that staff progress is currently tracked through paper timesheets collected at the end

	r Interview	status of an order as it is being picked and packed.	of the day. Therefore staff would need to be able to update the status of the order on the system.
Store Operations Staff	Problem Statement	The system must display real-time stock levels and product inventory to store operations staff.	The problem statement identifies that staff manually record and manage product stocks, which increases human error.
Store Operations Staff	Shift Manager Stakeholder Interview	The system must send a notification to store operations staff when a new order is assigned and requires picking and packing.	The shift manager stated that staff are currently informed through paper receipts on a physical board with no digital notification system.
Shift Manager	Shift Manager Stakeholder Interview	The system must allow shift managers to assign tasks to store operations staff.	Stated from the shift manager that tasks are currently assigned through daily in-person meetings, which is time-consuming and makes it difficult to organise efficiently.
Shift Manager	Shift Manager Stakeholder Interview	The system must allow shift managers to monitor the progress of assigned tasks in real time through a dashboard.	The shift manager stakeholder wanted a dashboard to view all store operations staff, their tasks, and the time spent on completing them.
Shift Manager	Shift Manager Stakeholder Interview	The system must send a notification to the shift manager when an assigned task has not been updated within a defined time period.	The stakeholder stated that it is challenging to notify the operation staff employees due to needing to meet in person to inform them thus indicating a need for a notification system.

Shift Manager	Shift Manager Stakeholder Interview	The system must allow shift managers to reassign tasks to different staff members when needed.	The shift manager stakeholder stated that when a staff member falls behind on their tasks, additional staff help would be assigned to that task. The system should support this by allowing shift managers to reassign tasks digitally.
Store Manager	Problem Statement	The system must allow the store manager to track the orders received.	From the problem statement, the store manager currently relies on manual reporting, which could lead to errors and is inefficient, indicating that the system would need to efficiently track orders received.
Store Manager	Problem Statement	The system must allow the store manager to view the orders that are being processed.	The problem statement identifies that store managers have no central system to track the performance of the store, making it difficult to determine potential operational bottlenecks. This means that the system would need to be able to view orders so that the performance of the store can be tracked.
Store Manager	Problem Statement	The system must allow the store manager to view the fulfilled orders.	The problem statement identifies that store managers rely on manual reporting, which makes it difficult to efficiently track store performance, indicating a need for a digital system to view fulfilled orders.
Store Manager	Problem Statement	The system must allow the store manager to be able to view the stock quantities for each product in the store.	In the problem statement there stated that manual reporting could have human errors, which could have major consequences of incorrect reporting of stock quantities.

Store Manager	Problem Statement	The system must allow the store manager to review customer feedback that is submitted through the system.	The problem statement identifies that customer service has no organised method to manage complaints and issues, indicating a need for the store manager to be able to review customer feedback digitally with the customer service representative to devise potential solutions.
Store Manager	Problem Statement	The system must allow the store manager to update in-store information on the online platform, which includes online promotions, announcements, and opening hours.	Since there is no online platform identified in the problem statement currently, having store information online would help inform customers and attract more customers.
Procurement Specialist	Procurement Specialist Stakeholder Interview	The system must allow the procurement specialist to generate reports, including current stock levels, product demand trends, and outstanding purchase orders.	In the procurement specialist stakeholder interview question is asked about what information would be helpful to make better purchasing decisions. Therefore, generating a report based on current stock levels, demand trends, and outstanding purchase orders would help with deciding purchase decisions.
Procurement Specialist	Procurement Specialist Stakeholder Interview	The system must notify the procurement specialist on the online platform when the stock is below a defined threshold level. The notification must apply to all	When asked about questions on how the procurement specialist stakeholder determines when to restock on inventory, it is tracked manually on a spreadsheet on a daily basis, making it difficult to respond to low stock in a timely manner across multiple store

		the store locations that the procurement specialist is assigned to.	locations.
Procurement Specialist	Procurement Specialist Stakeholder Interview	The system must allow the procurement specialist to create and submit purchase orders to suppliers and track their delivery status until fulfillment.	Stated from the inquiry about tracking suppliers fulfillment. This is done by inquiring with the supplier about their progress in fulfilling the order and tracking them on a spreadsheet to be checked manually on a daily basis.
Procurement Specialist	Procurement Specialist Stakeholder Interview	The system must allow the procurement specialist to view product inventory.	Real-time access to inventory is one of the key pieces of information needed to make better purchasing decisions, as stock is currently forecast based on historical data rather than real-time visibility.
Procurement Specialist	Procurement Specialist Stakeholder Interview	The system must provide procurement specialists with access to supplier market and pricing data, enabling them to compare suppliers based on cost, quality standards, and reliability.	Identifying the most suitable supplier for budget and standards is the biggest challenge, as supplier research is currently done manually with no system to compare suppliers efficiently.
Inventory Controller	Problem Statement	The system must allow the inventory controller to view product inventory across assigned store locations on the online platform.	Inventory controllers rely on procurement specialists' spreadsheet data, which is inaccurate and not real-time, risking overstocking or understocking across various store locations.

Inventory Controller	Problem Statement & Procurement Specialist Stakeholder Interview	The system must allow the inventory controller to monitor incoming stocks of suppliers, automatically updating inventory records upon delivery confirmation to ensure stock counts remain accurate across all locations.	Inventory controllers maintain optimal stock levels based on the procurement specialist's data, which is currently shared through a spreadsheet. The procurement specialist stakeholder stated that stock information is written on a spreadsheet and shared with the inventory controller to manage incoming stock, indicating a need for an automated system to update inventory records accurately.
Inventory Controller	Problem Statement	The system must allow inventory controllers to generate inventory reports that rank products by restocking frequency and current stock levels.	In the problem statement, inventory controllers monitor and maintain optimal stock levels across store locations, which risks overstocking or understocking without accurate data. A reporting system would allow inventory controllers to identify which products need restocking the most and the least to make informed stock management decisions.
Delivery Employee	Problem Statement	The system must automatically generate a delivery schedule based on the delivery driver's location and availability.	The problem statement identifies that there is no structured way to receive or manage deliveries, needing to manually create a schedule for delivery drivers. Thus, automatically generating schedules would greatly help in providing a smoother operation.
Delivery Employee	Problem Statement & Customer Stakeholder	The system must allow the delivery driver to update their status to dispatched, in transit, and delivered.	Delivery drivers have no efficient way to communicate delivery progress. The customer stakeholder stated they prefer in-app notifications and SMS updates

	r Interview		about their delivery status, which requires drivers to update their delivery status in real time through the system.
Delivery Employee	Problem Statement	The system must allow the delivery driver to flag urgent issues on the system if a delivery has any issues, such as failed deliveries, incorrect stock, or vehicle problems. This issue is then notified to the appropriate department.	Due to the problem statement, identifying that there is no indication of the delivery driver's progress or issues developed during delivery. This would leave customers uninformed about their order status while also preventing relevant departments from being notified of urgent delivery issues, impacting the efficiency and satisfaction of both the customer and the operation.

Requirement UML Modelling

In this subsystem, it is split into two use case diagrams; one use case diagram is to display the use case of the order and delivery online platform that includes customers and customer service representatives. The other use case diagram is to display the internal management system, where the use cases are about the operations that are performed internally on the online system. The actors include store operation staff, shift manager, store manager, procurement specialist, inventory controller, and delivery employee.

Use Case Descriptions

Field	Content
ID	Co1
Name	Search for Products
Purpose	Narrow down the products viewed so that the user can easily find what they are looking for.
Description	This use case allows customers to search for products on the online store. The user would enter the product name to be able to view all the products that relate to that name.
Actors	Customer
Data Content	The customer gives a search term to the system, and the system allows the customer to view the products offered.
Preconditions	Enter a search term
Trigger	Submitting search item
Event Flow	1. The customer goes to the main product viewing page. 2. Customer types in the search bar. 3. Customer submits the search term. 4. System retrieves and displays the product that matches or relates to the search term. 5. The customer could optionally select the filter functionality.
Ends When	It ends when the customer finishes typing the search term.
Include	None
Extension	Apply filter

Exceptions	If no product matches the search query, the system would show no products.
Post Conditions	A list of products matching the search query is displayed to the customer.
References	• User requirement (Browse and search for products online.) • Problem Statement (There is no efficient way in which customers are able to compare and decide different product choices based on their nutritional value, price, and other product attributes.) • Functional requirement (The system must allow customers to browse and search for products by name, category, price, and nutritional value.)

Field	Content
ID	Co2
Name	Place Order
Purpose	This use case is the final step to ordering the products.
Description	The customer reviews their shopping cart, selects their delivery time and destination, and submits their order. The system records the order for the store operation staff for picking and packing.
Actors	Customer
Data Content	This use case requires the customer account details, shopping cart details, and delivery time and destination.
Preconditions	• The customer must log in to their account. • They must have at least 1 product in their cart.
Trigger	Initiates checkout from their cart.
Event Flow	1. Customer proceeds to checkout from their cart. 2. System displays their order. 3. Confirm the order details. 4. Select delivery time and destination. 5. Submit order.

	6. System records orders. 7. The system notifies store operations staff of the new order requiring picking and packing.
Ends When	The order is successfully submitted and recorded in the system.
Include	Select delivery information
Extension	None
Exceptions	If a product in the cart is out of stock at the time of submission, the system notifies the customer and removes the unavailable item before confirming the order.
Post Conditions	• The order is recorded in the system. • Store operations are informed about the new order for picking and packing.
References	• User requirement (Add products to a shopping cart and check out.) • Functional requirement (The system must allow customers to proceed to checkout from their shopping cart, where they can review their orders.)

Field	Content
ID	CSR01
Name	Resolve Complaint
Purpose	To allow the customer service representative to resolve a customer complaint and notify the customer of the outcome.
Description	This provides customer service to investigate the issue and potentially resolve it. This also extends to the use case refund if an issue needs to be refunded to

	the customer, as well as notifying the customer of the status of the issue.
Actors	Customer Service Representative, Customer
Data Content	This use case requires the customer complaint details, order details related to the complaint, solution details, Refund details if needed, and notification details sent to the customer.
Preconditions	• A complaint is submitted by the customer. • Complaint status is tracked and updated before resolving.
Trigger	When the customer service representative determined that the complaint was fully solved.
Event Flow	1. The customer service representative opens a complaint. 2. The system displays complaint details. 3. Reviews the complaint and determines the solution. 4. Update the complaint on the current status for resolving the complaint. 5. The customer service representative marks the complaint as resolved. 6. Notify the customer of the complaint outcome.
Ends When	The complaint is resolved, and the customer is notified of the outcome.
Include	Send notification to customers
Extension	Process refund
Exceptions	• If a complaint needed for detail, the customer is contacted before resolving.
Post Conditions	• Complaint is marked is resolved. • Customer is notified of the complaint outcome. • If a refund is required, the customer is notified of the refund.
References	• Functional requirement (The system must allow customer service representatives to update and resolve logged complaints.)

	• Functional requirement (The system must allow customer service representatives to process refunds for incorrect or out-of-stock items.) • Functional requirement (The system must allow customer service representatives to send notifications to customers via in-app notifications or SMS regarding issues with their order.) • Problem statement (Deal with customer complaints, as well as having no organised method to manage digital customer complaints.)
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Field	Content
ID	SM01
Name	Assign Task
Purpose	To allow the shift manager to assign tasks to store operations staff and monitor their progress.
Description	This use case is to allow the shift manager to assign tasks in the system, monitor the progress of assigned tasks in real-time through a dashboard, and if a task is not completed and additional members are not available, store operation staff are able to be reassigned the task.
Actors	Shift Manager, Store Operations Staff
Data Content	Staff member details are needed, and task details (description and deadline), task status, progress update, reassignment detail, and notification detail for overdue tasks
Preconditions	• There must be available tasks for store operations staff to be assigned.
Trigger	The shift manager creates a task.
Event Flow	1. The system displays the list of available store operation staff and their current task status. 2. The shift manager selects a task and assign to the store operations staff.

	3. The system records the task assigned and notifies the store operation staff of the new task. 4. Tasks assigned are updated by store operations staff. 5. Store operation staff updates progress. 6. The shift manager monitors the progress of all assigned tasks through the dashboard in real time 7. The system sends a notification to the shift manager if a task has not been updated within a defined time period. 8. The shift manager reviews the notification and determines whether reassignment is necessary.
Ends When	All assigned task are completed.
Include	Monitor progress
Extension	Reassign task
Exceptions	● If no store operations staff are available to be assigned a task, the system notifies the shift manager and requires the shift manager to choose another staff member.
Post Conditions	● Tasks have been successfully assigned to store operations staff. ● Store operations staff have been notified of their assigned tasks. ● The shift manager can monitor all task progress through the dashboard in real time.
References	● Functional Requirement (The system must allow shift managers to assign tasks to store operations staff.) ● Functional Requirement (The system must allow shift managers to monitor the progress of assigned tasks in real time through a dashboard.) ● Functional Requirement (The system must send a notification to the shift manager when an assigned task has not been updated within a defined time period.)

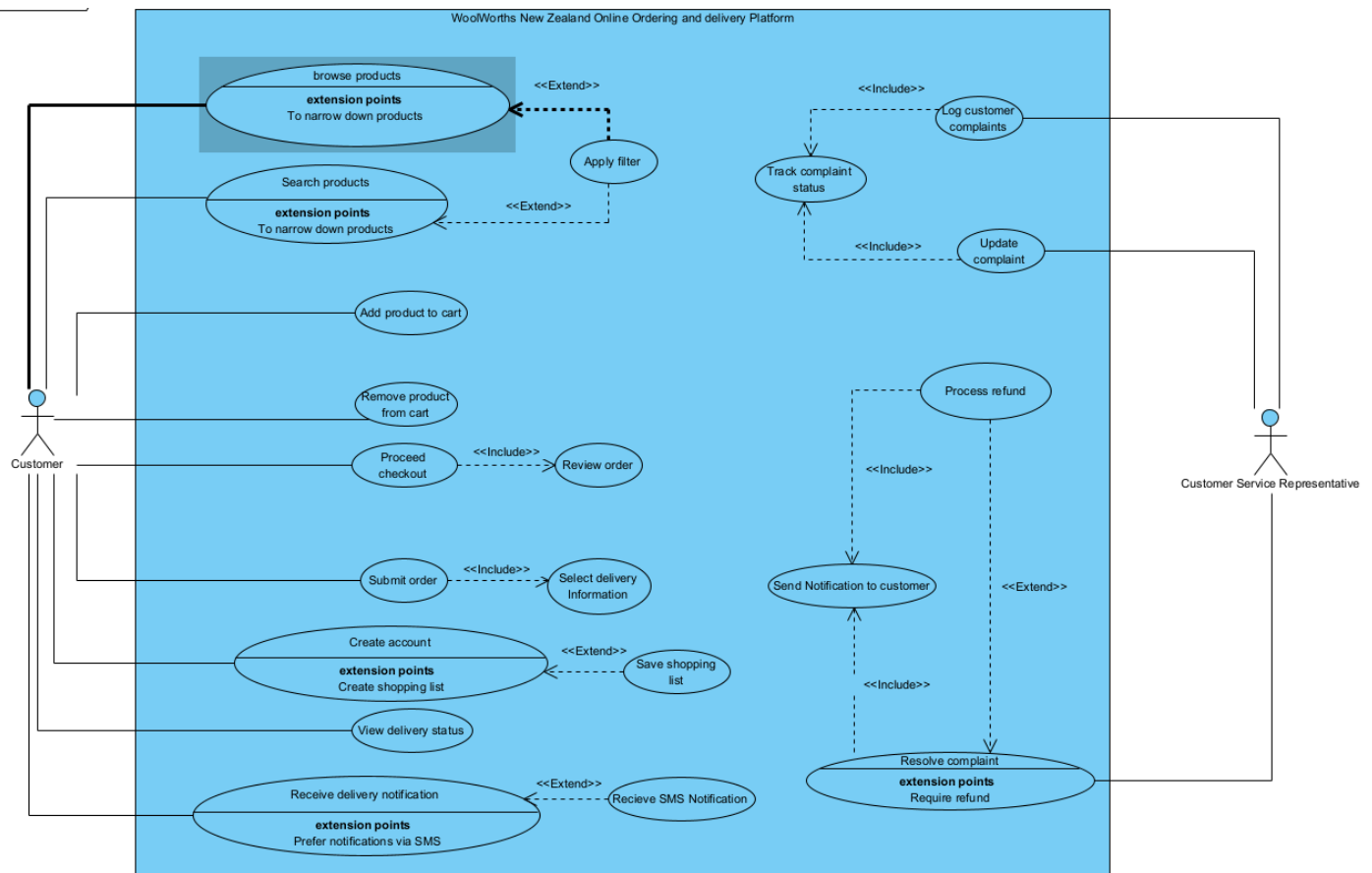
	• Functional Requirement (The system must allow shift managers to reassign tasks to different staff members when needed.) • Problem Statement (Shift managers have no digital tools to delegate tasks or monitor the progress of staff, relying entirely on in-person communication.)
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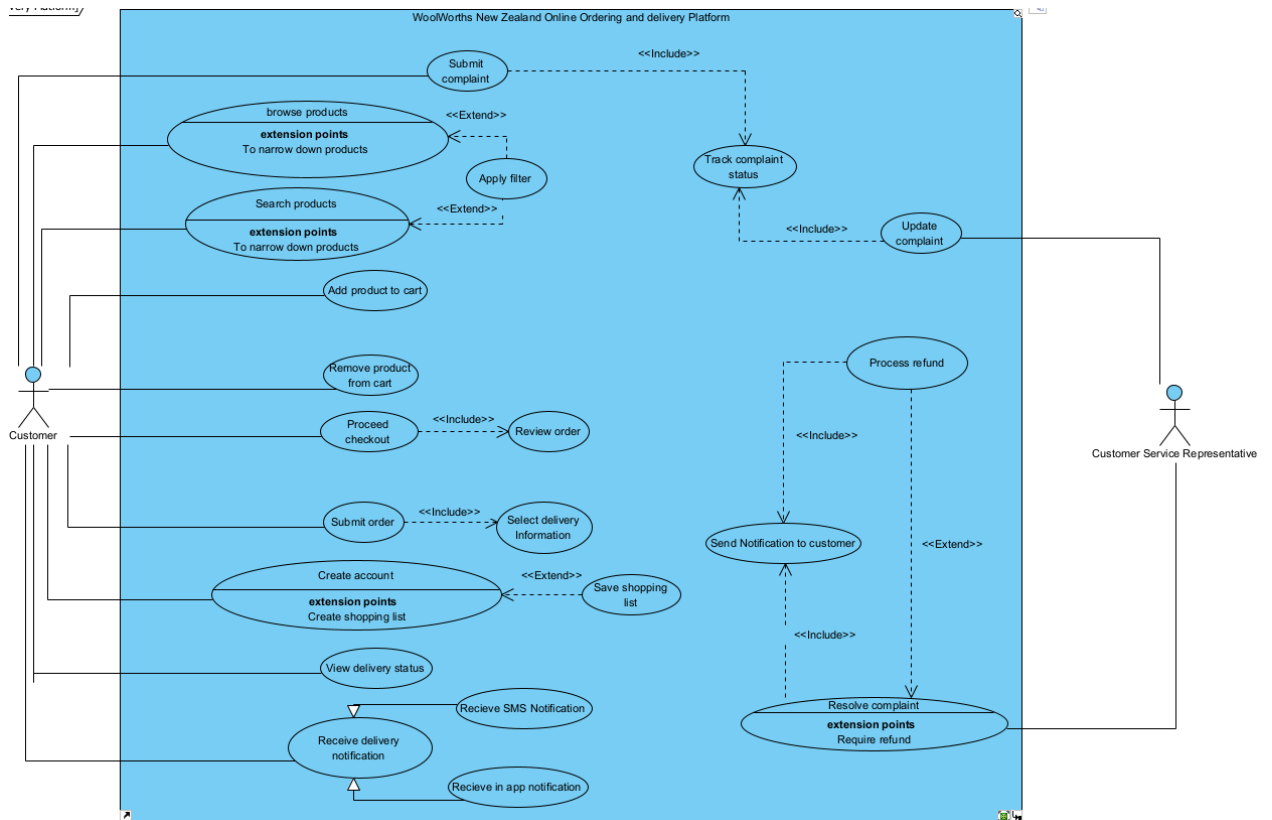
Field	Content
ID	PSO1
Name	Create and Submit Purchase Order
Purpose	Provide the procurement specialist to create and submit a purchase order to a supplier and track its delivery status until fulfilment.
Description	The procurement specialist reviews current stock levels and supplier data to determine what products need to be ordered. They create and submit a purchase order to the appropriate supplier through the system. The system tracks the delivery status of the purchase order until it is fulfilled. If a low stock alert is received, it may trigger the procurement specialist to create a new purchase order.
Actors	Procurement Specialist
Data Content	Product details (name, quantity), supplier details (name, pricing reliability), purchase order details (order date, delivery date), delivery status update from supplier, and low stock alert details.
Preconditions	• A procurement specialist must be assigned to one or more store locations. • Supplier data is available on the system.
Trigger	The procurement specialist identified that a product needs to be restocked.
Event Flow	1. System displays current stock level and produces demand trends

	across assigned store locations. 2. Review stock level and identify which needs to be restocked. 3. Select the appropriate supplier. 4. Creates new purchase order. 5. Submit the purchase order to the supplier through the system. 6. System records purchase orders and tracks delivery status. 7. System update delivery status based on supplier fulfillment. 8. The inventory controller is notified when stock is delivered.
Ends When	The order purchase has been fulfilled by the supplier, and the inventory stock is updated.
Include	Track supplier fulfillment
Extension	Create and Submit Purchase Order extends Receive Low Stock Alert
Exceptions	• If the selected supplier is unavailable or cannot fulfil the order, the system notifies the procurement specialist and prompts them to select an alternative supplier.
Post Conditions	• The purchase order is recorded on the system. • The delivery status of the purchase is tracked by the system. • The inventory controller is notified of the new stock.
References	• Functional requirement (The system must allow the procurement specialist to create and submit purchase orders to suppliers and track their delivery status until fulfillment.) • Functional requirement (The system must notify the procurement specialist on the online platform when the stock is below a defined threshold level.) • Functional requirement (The system must provide procurement specialists with access to supplier market and pricing data, enabling them to compare suppliers based on cost, quality standards, and reliability.) • Problem statement (procurement specialists are unable to make the

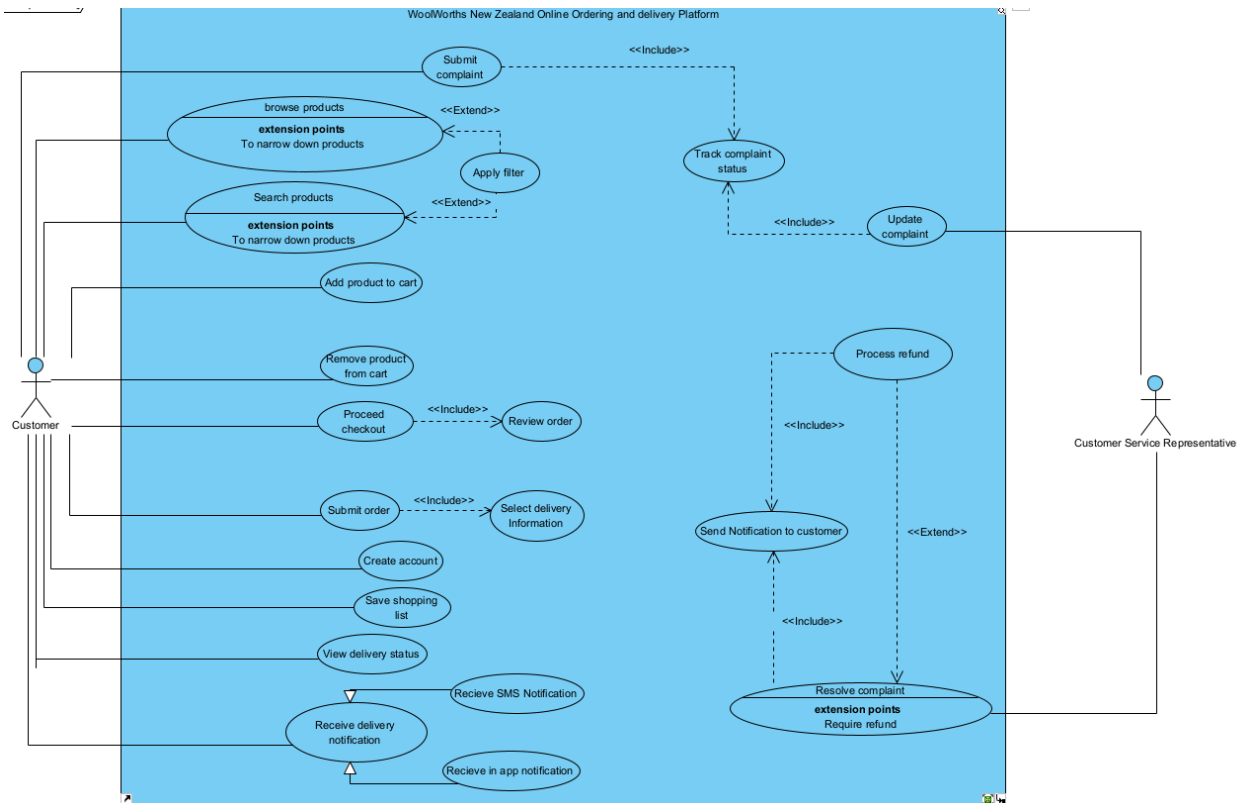
	most effective decisions in purchasing and sourcing products due to a lack of live visibility on real-time demand data.)
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Use Case Diagram (Online Ordering and Delivery Platform)

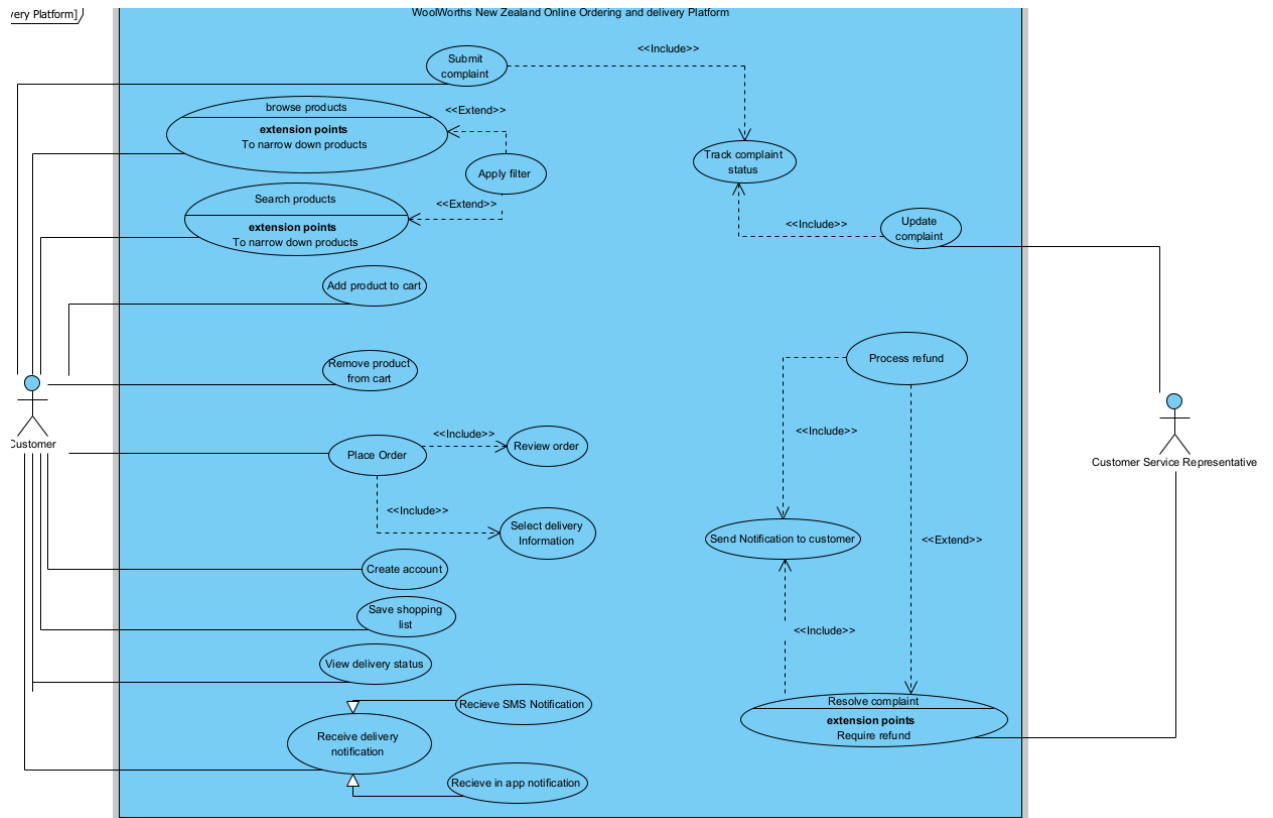




2nd iteration, added submit complaint as a use case for the customer, and included the track complaint status use case to it; therefore, the log complaint use case for the customer service representative is no longer required. Also added generalization to receive delivery notification, as receive SMS notification and receive in-app notification inherit receive delivery notification.

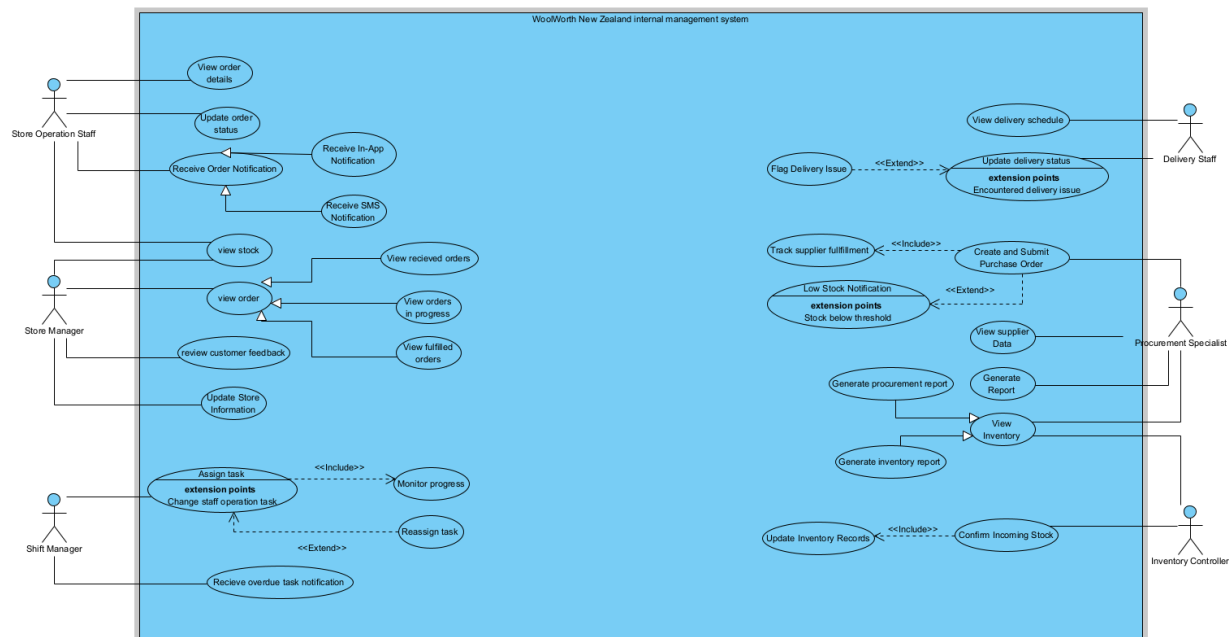


For the 3rd iteration, firstly, removed the extension from the create account, as customers could save a shopping list at any point, not just when the account is created.



The 4th iteration removed the proceed checkout and submit order since they are both very similar, therefore combined them both together as “place order” which includes both review order and select delivery information.

Use Case Diagram (Internal Management System)



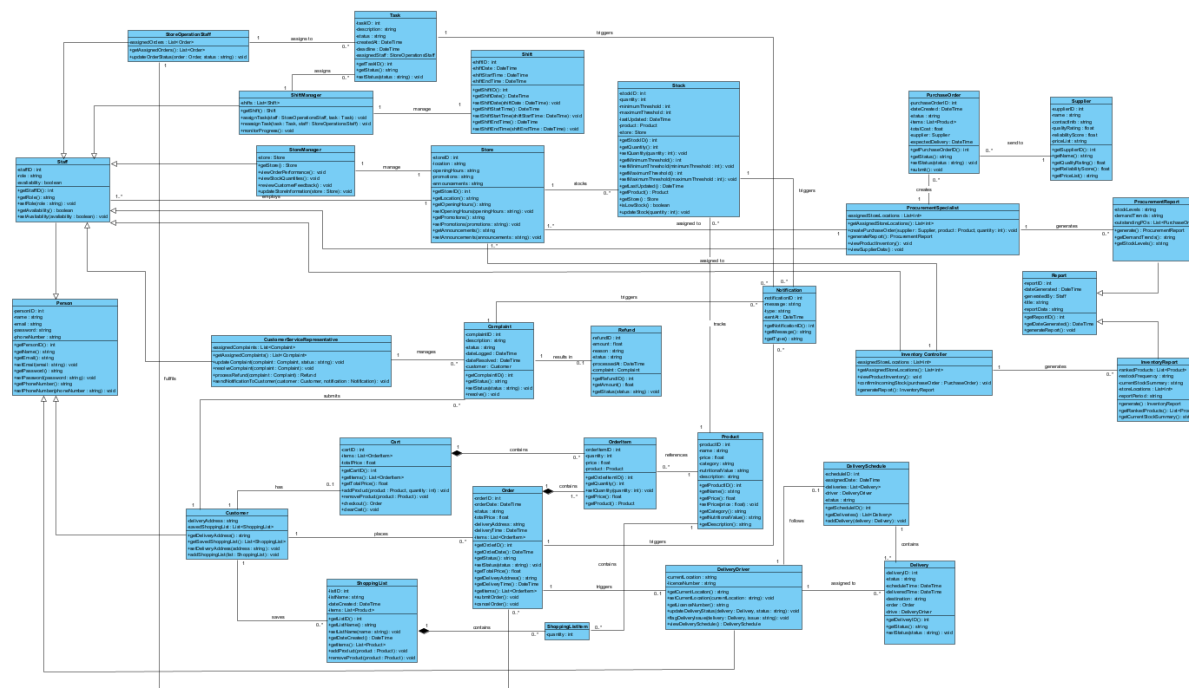
Initial Structural Design and Implementation

- ☒ Initial UML Class Diagram
- ☐ Mapping between UML classes and planned C# classes
- ☒ Identification of anticipated design weaknesses prior to OOP application
- ☐ OOP-based implementation in C# that strictly conforms to the UML models.
- ☒ In the Class diagram, provide reasons for each component by writing a short summary of each.

Class	Explanation
Person	The Person class is the base class from which the staff class and customer class inherit
Staff	The staff class contains the attributes that a staff member would need
Customer	The customer class contains the attributes that a customer would need
Store Operation Staff	This class inherits attributes from the staff class and contains an attribute that stores orders that are assigned to track the status
Shift Manager	Assigns tasks and staff to shifts, monitors progress
Store Manager	Manages store operations, orders, and stock
Customer Service Representative	Handles complaints, refunds, and customer notifications
Inventory Controller	Monitors stock, creates purchase orders, and generates inventory reports
Procurement Specialist	Manages supplier relationships and purchase orders
Store	Holds store info (location, hours, promotions, announcements)
Shift	Tracks shift times, status, and assigned staff
Stock	Tracks product quantity, thresholds, and restock needs

Cart	Holds items before checkout, calculates total price
ShoppingList	Saved product lists belonging to a customer
Shopping List Item	Individual items and quantities within a shopping list
Order	Tracks a placed order (date, status, address, items)
OrderItem	A single product line within an order (quantity, price)
Delivery Driver	Tracks driver location, license, and schedule
Delivery Schedule	Plans delivery times and links drivers to deliveries
Delivery	Represents the actual delivery (status, times, destination)
Product	Core product info (name, category, price, nutrition, description)
Supplier	Supplier contact details and reliability score
Purchase Order	Order placed with a supplier for restocking
Complaint	Customer complaint with status, resolution, and refund link
Refund	Tracks refund amount, status, and processing date
Notification	System messages sent to staff or customers
Report	The general report contains attributes that the procurement report and inventory report inherit
Inventory Report	Generate a report of stock levels and product locations
ProcurementReport	Summarizes stock levels and outstanding purchase orders

Class Diagram



Evaluation

Weaknesses

- Both cart and shoppingList hold a list of products with a similar structure. There is adding and removing products in both classes; therefore, to improve the design, the cart could temporarily be the shopping list, and you could have a save functionality that saves these cart details.
- The notification class handles all system messages with a string type. For the current prototype, this is sufficient, but in an actual implementation, there would need to be different notification types (email, SMS) that have different behaviors.
- The Report class is not an abstract class that can be initialized. The report class should be an abstract class to force the report to be either a procurement report or an inventory report.

- I have no abstract or interface used in the class diagram, such as person should be an abstract class cause it shouldn't be able to be initialized, as well as no interface that would enforce consistent methods across classes. This makes the system less flexible, maintainable, and scalable.

Limitations

- Even though I stated that the payment functionality is handled through a third-party gateway. Without it, there's no way for customers to actually purchase products, track payment, status, or history. Thus, this prototype would not be functional, as this design implementation does not take into account the payment.
- The system handles refunds but has no dedicated Return or Exchange class. A customer may exchange without needing a refund; therefore, an extra class would be needed to include this functionality.
- There's no class that tracks the changes of data over time, such as who changed the order status, when stock is updated, and who processed the refund. This would be important in making sure that there is accountability.
- In the staff class, the availability attribute is not enough to represent when the staff is available. Other details that are important include the working days, part-time hours, holiday leave, and sick days. For a smooth operation they details would need to be taken into account for planning shifts of employees.

Risk

- The DeliveryDriver store location should be based on latitude/longitude instead of the location as a string. Storing it as a string would not allow the system to calculate the distance between the delivery driver and the delivery destination.

- The minimum stock threshold and the maximum threshold attributes should be dependent on the market trend; this is more difficult to measure, but it is more reliable for determining the amount of stock needed. This could result in the procurement specialist ordering too much stock or running out at critical times, both of which have financial consequences for the business.
- There is no sessionID attribute or time attribute in the customer, which is needed to track when the cart was created. If the customer leaves the system and re-enters the online order system at another point in time, the cart would not be able to retrieve the previously saved cart.

References

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Woolworths NZ Ltd.

https://www.woolworths.co.nz/info/policies/privacy-policy?promo_position=Tile&promo_creative=Tile&promo_name=OurPolicies&itemID=Privacy%20Policy

Appendices

- GitHub

(<https://github.com/ShawnLeeyz/Woolworths-Systems-Analysis-Design-and-Modelling-Project-I.git>)